

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Built a registry-governed semantic layer over BigQuery: YAML definitions plus SQLite vocabulary snapshots define legal dimensions, values, and table wiring; a deterministic prompt builder generates the LLM contract from the registry; a single LLM call maps the question to strict JSON intent (the model never writes SQL); a deterministic gate validates and canonicalizes intent into a typed dataclass before a deterministic compiler emits the SQL — same question, same answer, every time. New or changed definitions pass a human review gate before entering the registry.
- Translated ambiguous business questions into a formal, governed representation: 16 governed demographic dimensions, case-insensitive canonical matching with synonym lookup, UPC normalization to 13-digit warehouse form with optional live BigQuery existence checks, and a typed feasibility intent that boxes the only probabilistic step so it is regression-testable. Partnered with Product and Data Science to keep definitions accessible to non-technical stakeholders.
- Designed and shipped a self-service analytics agent over 67 BigQuery tables with sub-agent orchestration and context-isolated validation (Context Researcher → SQL Drafter → Validator → Devil's Advocate); 8 deterministic SQL rules and a 23-case golden-query evaluation suite. Approved by Sr. Director and product leadership and currently the only AI skill in active use by business teams on Data Ventures.
- Built an autonomous Jira ticket-resolution agent: 6-gate triage, MCP-driven context gathering, FAISS+BM25 retrieval over 11,000 historical SQL queries, self-healing execution with a 10-retry budget. Closed 400+ data requests across 8 categories over 15 months — cut turnaround from 30–60 minutes per request (up to a full day for complex ones) to under 10 minutes each.
- Replaced ~20 distinct recruitment workflows run by hand 1–2× weekly with end-to-end automation on a 68-table BigQuery platform integrating relational and semi-structured sources; stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates. A recurring 3-hour task now takes ~3 minutes of setup.
- Built a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; state persists to BigQuery so the agent has no LLM-memory reliance. Designed a 3-tier event-driven v2 self-configuring via a BigQuery event registry — new event types ship without code deploys.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Semantic modeling & governance: registry-governed semantic layer (YAML definitions + SQLite vocabulary snapshots), controlled vocabularies, canonical/synonym resolution, typed intent contracts, deterministic validation, human review gate on definition changes, versioned definitions

Data: BigQuery, dbt, SQL, Python, PySpark, Airflow, Databricks, MySQL

Agent stack: MCP server design, sub-agent orchestration, context engineering, prompt engineering, RAG (FAISS + BM25 + RRF), evaluation suites (golden sets, human-in-the-loop validation), headless cron agents, event-driven architectures, decision logging / observability

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

APIs / triggers / glue: FastAPI, Cloud Run, BigQuery scheduled queries, Slack webhooks, cron, Playwright SSO, JSONL state stores

Viz: Power BI (DAX, XSLT), Tableau, LookML, Streamlit, Matplotlib, React Recharts

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL